MATH 345 Skeleton Notes

Unit 5: Eigenvalues and second-order optimization

Section 5.1: Eigenvalues, eigenvectors, and diagonalization

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Let A be a square matrix. An eigenvector is a nonzero vector whose line is preserved by A: applying A may stretch, reverse, or collapse the vector, but it does not turn the vector away from that line. Eigenvectors are general tools for understanding matrix actions, independently of any particular application.

Basic definitions

Definition

Let A be an $n \times n$ matrix. Suppose $\mathbf{v}$ is a *nonzero* vector in $\mathbb{R}^n$ and λ is a scalar (which could be 0) such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Then λ is an eigenvalue of A, and $\mathbf{v}$ an eigenvector of A associated with the eigenvalue λ.

Activity: Checking eigenvectors and their eigenvalues (U5-LO1)

Check that

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

are eigenvectors for the matrix

$$A = \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix},$$

and find the corresponding eigenvalues.

Activity: Finding eigenvalues and eigenvectors (U5-LO1)

Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$$

be a matrix. Find the eigenvalues and eigenvectors of A.

Definition

Let $A = [a_{ij}]$ be an $n \times n$ matrix. Then

$$\det(\lambda I_n - A) = \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix}$$

is called the **characteristic polynomial** of A and is sometimes denoted $c_A(\lambda)$. The equation

$$\det(\lambda I_n - A) = 0$$

is called the **characteristic equation** of A.

Remark

Some authors define the characteristic polynomial to be $\det(A - \lambda I_n)$ rather than $\det(\lambda I_n - A)$.

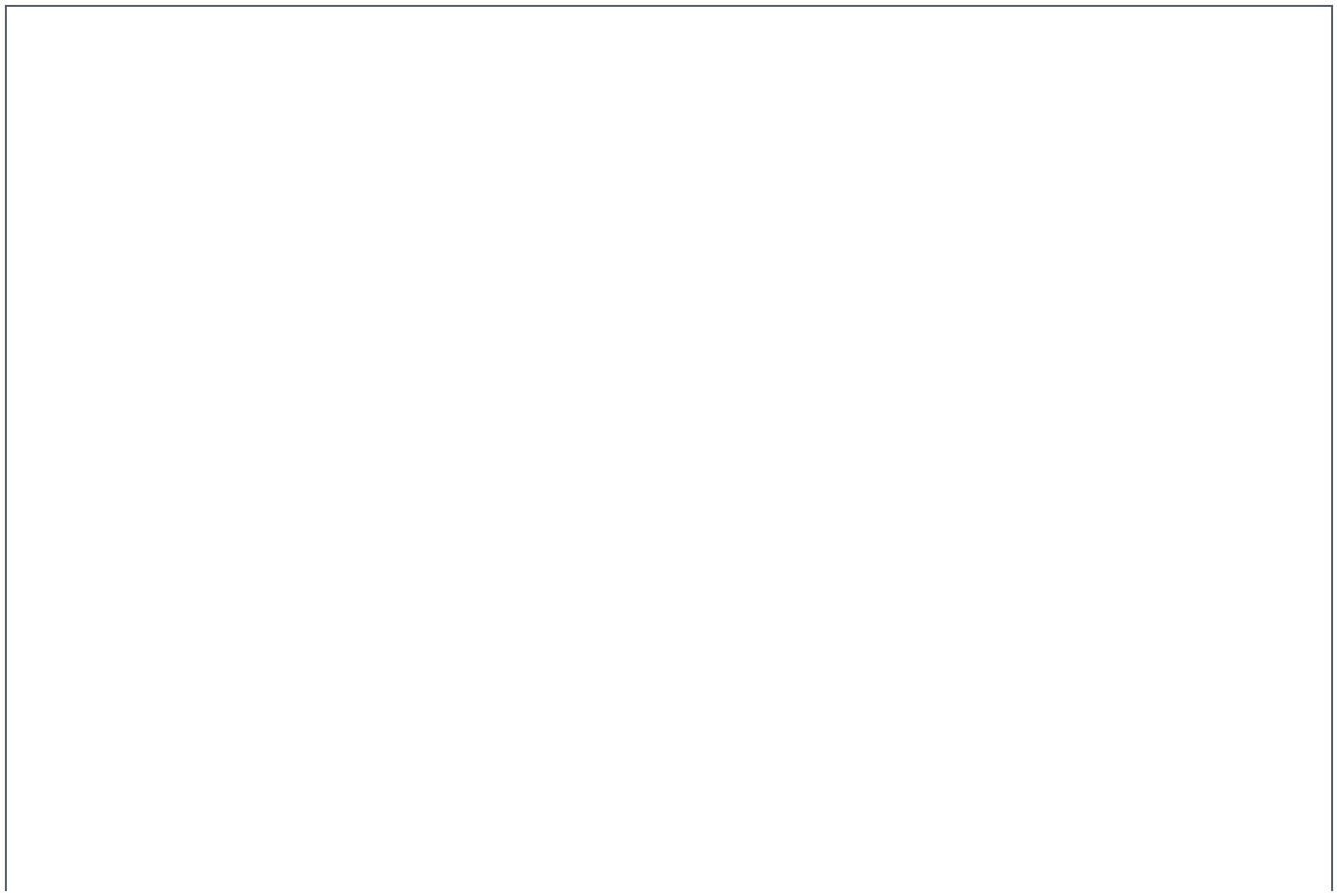
Theorem

Let A be an $n \times n$ matrix. The eigenvalues of A are the *roots* of the characteristic polynomial of A.
Recall that a root of a polynomial p is a value r such that $p(r) = 0$.

Activity: Computing a characteristic polynomial (U5-LO1)

Find the characteristic polynomial of A if

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}.$$



Note: Finding eigenvalues and eigenvectors of a matrix A

1. Determine the roots of the characteristic polynomial $c_A(\lambda) = \det(\lambda I_n - A)$. These are the *eigenvalues* of A.

2. For each eigenvalue λ, find [all] nontrivial solutions to the homogeneous system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$. These are the *eigenvectors* of A associated with λ.

Definition

A basic solution to the homogeneous equation $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ will be called a [basic eigenvector] associated with the eigenvalue λ.

Warning

A common mistake is to first row-reduce A to RREF and then find the eigenvalues and eigenvectors of the RREF matrix. Row operations on the matrix A *do not* preserve the set of basic solutions. They only preserve solutions to the homogeneous equation $A\mathbf{v} = \mathbf{0}$.

If $\mathbf{v}$ is an eigenvector of a matrix A with eigenvalue λ, then any nonzero scalar multiple of $\mathbf{v}$ is also an eigenvector with the same eigenvalue λ. Indeed, if $\mathbf{w} = c\mathbf{v}$ for some scalar $c \neq 0$, then:

$$\begin{aligned} A\mathbf{w} &= A(c\mathbf{v}) \\ &= c(A\mathbf{v}) \quad \text{(by linearity of matrix multiplication)} \\ &= c(\lambda\mathbf{v}) \quad \text{(since } \mathbf{v} \text{ is an eigenvector with eigenvalue } \lambda) \\ &= \lambda(c\mathbf{v}) \\ &= \lambda\mathbf{w} \end{aligned}$$

Put differently, the entire line span($\mathbf{v}$) is "invariant" under the map defined by A, with every vector in this line being stretched (or compressed) by the factor λ.

Definition: Eigenspaces of a matrix

Suppose A is a square matrix. Then for any scalar λ, the set

$$E_\lambda(A) = \{\mathbf{x} \in \mathbb{R}^n \ : \ A\mathbf{x} = \lambda\mathbf{x}\}$$

is called the eigenspace of A corresponding to λ.

Theorem

Suppose A is a square matrix and $\lambda \in \mathbb{R}$. Then $E_\lambda(A)$ is a subspace of $\mathbb{R}^n$.

Why is this true?

Note that $\mathbf{x} \in E_\lambda(A)$ precisely when $\mathbf{x}$ solves the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$. Thus $E_\lambda(A)$ is precisely the null space of the matrix $A - \lambda I$, and the theorem follows from Null and image spaces are subspaces.

Activity: Eigenvalues and eigenspace bases (U5-LO1)

Find the eigenvalues and basic eigenvectors of the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Diagonalization

Eigenvalues describe individual special directions. Diagonalization asks whether there are enough independent eigenvectors to form a basis. In eigenvector coordinates, the matrix acts by scaling each coordinate separately.

Definition

An $n \times n$ matrix A is called diagonalizable if there exists an invertible $n \times n$ matrix P so that $P^{-1}AP$ is a diagonal matrix (recall Triangular and Diagonal Matrices). The matrix P is called a diagonalizing matrix for A.

Activity: Diagonalization and a matrix power (U5-LO2)

Suppose

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}.$$

Compute $P^{-1}AP$ where

$$P = \begin{bmatrix} -1/2 & -1 \\ 1 & 1 \end{bmatrix}.$$

You may use the fact that

$$P^{-1} = \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix}.$$

What does this computation say about the matrix A?

Activity: Diagonalization and a matrix power (U5-LO2)

Suppose

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}.$$

Use the previous computation to calculate A^5.

The previous activity shows why diagonalizing a matrix may be useful. But given a matrix A, how do we potentially find P such that $P^{-1}AP$ is a diagonal matrix?

Activity: Interpreting the columns in a diagonalization (U5-LO1, U5-LO2)

Suppose $P^{-1}AP = D$ for some diagonal matrix $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$. Let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be the columns of P. Note that the equation $P^{-1}AP = D$ holds if and only if $AP = PD$.
What happens when I take PD?

Activity: Interpreting the columns in a diagonalization (U5-LO1, U5-LO2)

Suppose $P^{-1}AP = D$ for some diagonal matrix $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be the columns of P. Note that the equation $P^{-1}AP = D$ holds if and only if $AP = PD$.

Calculate AP and PD in terms of the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$, the scalars $\lambda_1, \ldots, \lambda_n$ and the matrix A.

Activity: Interpreting the columns in a diagonalization (U5-LO1, U5-LO2)

Suppose $P^{-1}AP = D$ for some diagonal matrix $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$. Let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be the columns of P. Note that the equation $P^{-1}AP = D$ holds if and only if $AP = PD$.
So what conditions on P and D are necessary and sufficient for the equation $AP = PD$ to hold?

To see this differently, suppose $P^{-1}AP$ is a diagonal matrix with diagonal

entries $\lambda_1, \ldots, \lambda_n$. Then for each standard basis vector (recall Standard basis) $P^{-1}AP\mathbf{e}_i = \lambda_i \mathbf{e}_i$. Multiplying this equation by P on the left-hand side and setting $\mathbf{v}_i = P\mathbf{e}_i$, we obtain

$$A\mathbf{v}_i = P(P^{-1}AP\mathbf{e}_i) = P(\lambda_i \mathbf{e}_i) = \lambda_i \mathbf{v}_i.$$

Thus $\mathbf{v}_i$ is an eigenvector of the matrix A associated with the eigenvalue λ_i.

Theorem: Diagonalizability criterion

An $n \times n$ matrix A is *diagonalizable* if and only if A has n eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ such that the matrix P with columns $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is invertible. In this case, $P^{-1}AP = D$, where D is a diagonal matrix such that the elements of D on the main diagonal are the *eigenvalues* of A.

Remark

Recalling Invertibility Criteria, the matrix P in Diagonalizability criterion is invertible if and only if the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ form a basis for $\mathbb{R}^n$.

If $\mathbb{R}^n$ has a basis of eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$, then every vector $\mathbf{v} \in \mathbb{R}^n$ can be written as a linear combination of these eigenvectors, i.e., there exists constants $a_1, \ldots, a_n$ such that

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n.$$

Thus

$$\begin{aligned} T_A(\mathbf{v}) &= A\mathbf{v} \\ &= A(a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n) \\ &= a_1A\mathbf{v}_1 + a_2A\mathbf{v}_2 + \cdots + a_nA\mathbf{v}_n \\ &= a_1\lambda_1\mathbf{v}_1 + a_2\lambda_2\mathbf{v}_2 + \cdots + a_n\lambda_n\mathbf{v}_n \\ &= \lambda_1(a_1\mathbf{v}_1) + \lambda_2(a_2\mathbf{v}_2) + \cdots + \lambda_n(a_n\mathbf{v}_n) \end{aligned}$$

In other words, T_A 'stretches' the components of $\mathbf{v}$ along each eigenvector by the corresponding eigenvalue.

Activity: Checking a proposed eigenvector pair (U5-LO1)

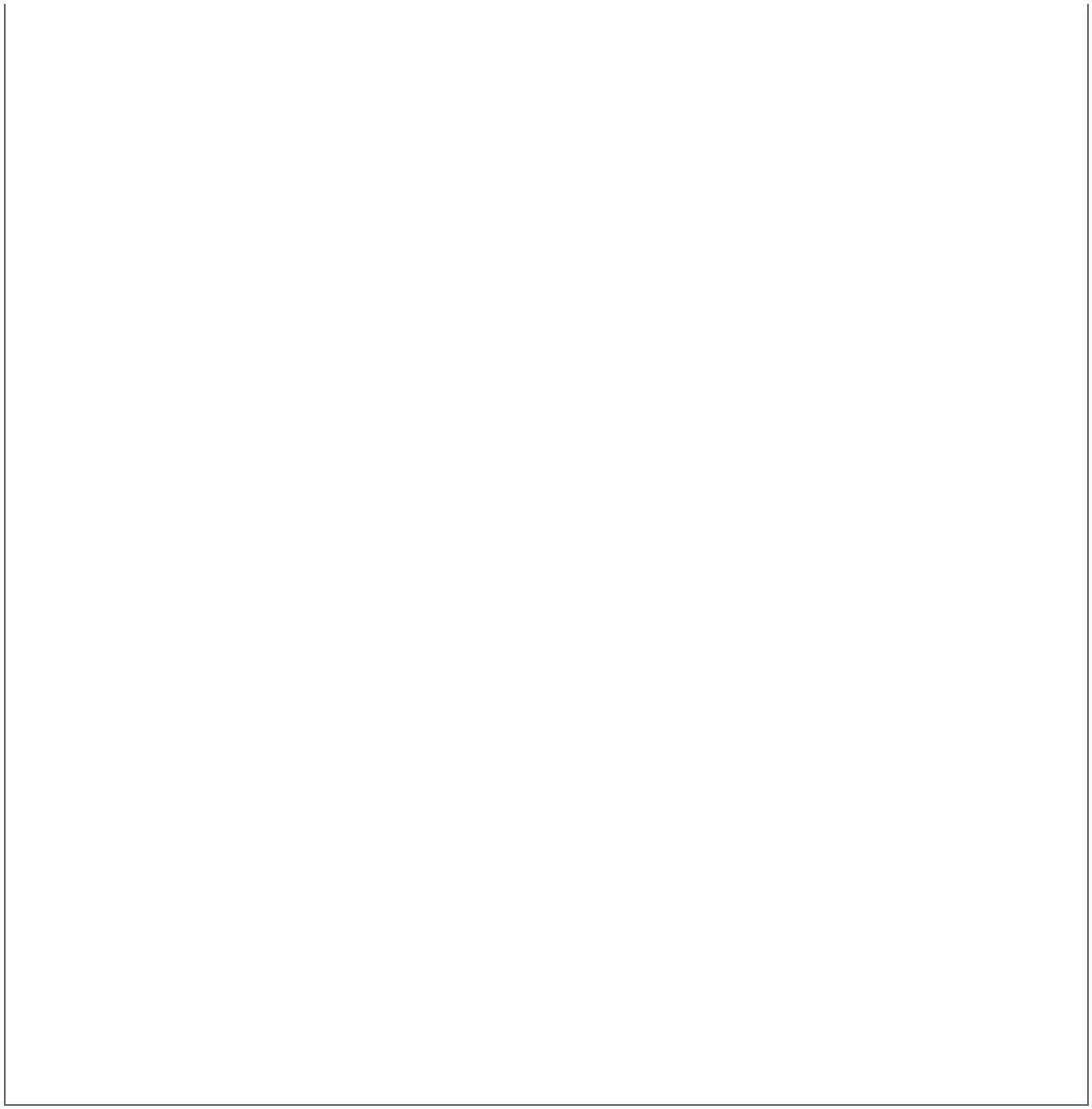
Consider the matrix

$$M = \begin{bmatrix} 5/3 & -2/3 \\ -1/3 & 4/3 \end{bmatrix}.$$

Verify that

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$$

are eigenvectors of M corresponding to different eigenvalues. What are the eigenvalues?



The figure illustrates the effect of the matrix M from Checking a proposed eigenvector pair (U5-LO1) on a grid in $\mathbb{R}^2$. In particular, it highlights an important geometric interpretation of eigenvectors and eigenvalues; when we apply the linear map defined by the matrix M to any vector in $\mathbb{R}^2$, the matrix map stretches eigenvector directions by their eigenvalues.

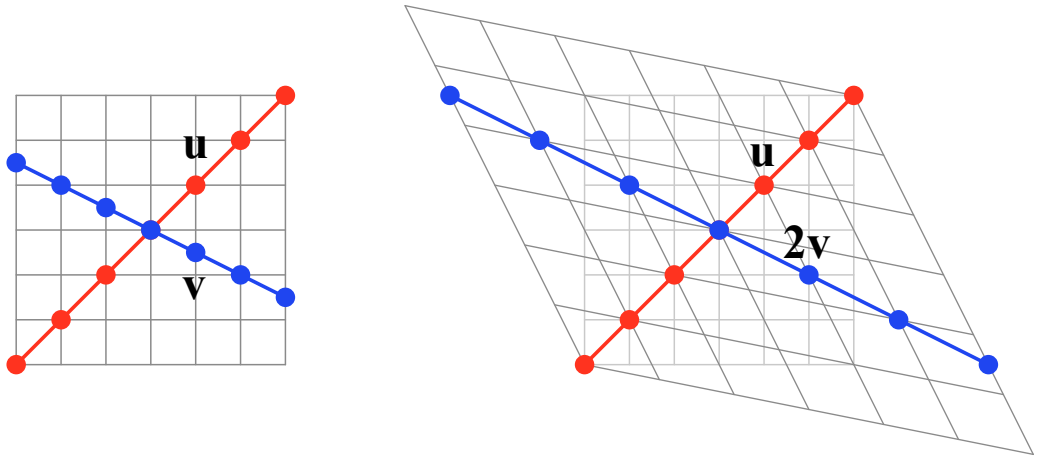


Figure. Before-and-after grid diagram for a matrix with two highlighted eigenvector directions.

Remark: Non-diagonalizable matrices

Consider the rotation matrix

$$A_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

where θ is not a multiple of $180°$. For any nonzero vector $\mathbf{v} \in \mathbb{R}^2$, the vector $A_\theta \mathbf{v}$ is obtained by rotating $\mathbf{v}$ counterclockwise by an angle θ, so $A_\theta \mathbf{v}$ is never on the line spanned by $\mathbf{v}$. Thus $A_\theta \mathbf{v} = \lambda \mathbf{v}$ cannot hold for any real scalar λ and any nonzero $\mathbf{v}$.
Thus the matrix A_θ has no real eigenvalues or real eigenvectors. It is therefore not diagonalizable over $\mathbb{R}$. This makes geometric sense: we cannot find special directions in $\mathbb{R}^2$ so that rotating vectors in the plane is obtained by stretching in those directions.

Non-diagonalizable matrices illustrates why not every matrix is diagonalizable. A matrix is diagonalizable precisely when it has a basis of eigenvectors, but this is not always possible.

Activity: Testing a matrix for diagonalizability (U5-LO2)

Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}.$$

Is A diagonalizable?

Activity: A second diagonalizability test (U5-LO2)

Let

$$A = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}.$$

Is A diagonalizable?

